

Data Structure(CSL202)

Dr. Sk Subidh Ali (subidh@iitbhilai.ac.in)

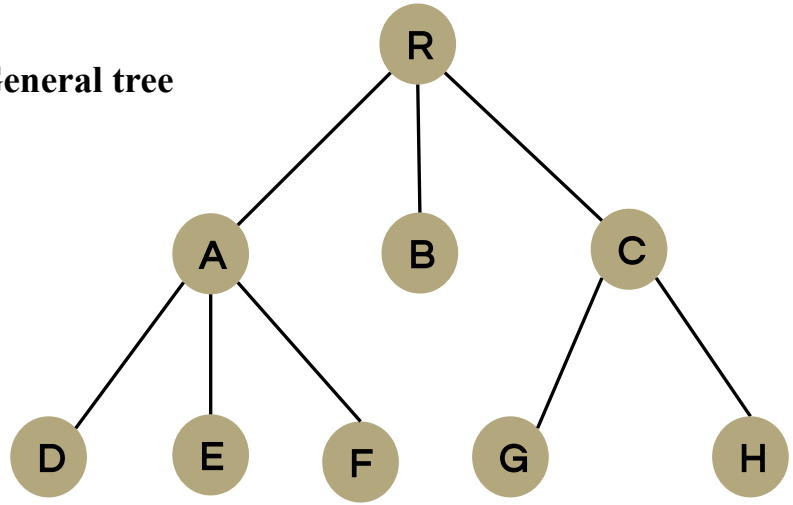

भारतीय प्रौद्योगिकी संस्थान भिलाई
Indian Institute of Technology Bhilai

Trees

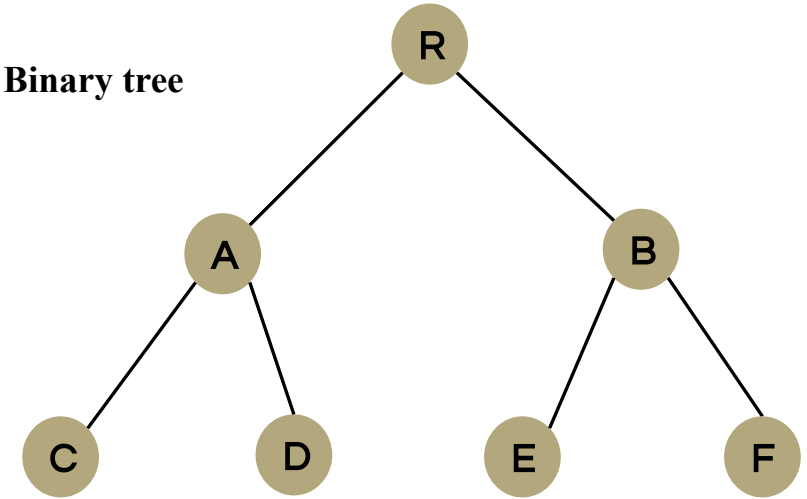
Trees

- Non-linear or hierarchical data structure.
- Collection of elements called “nodes”.

General tree



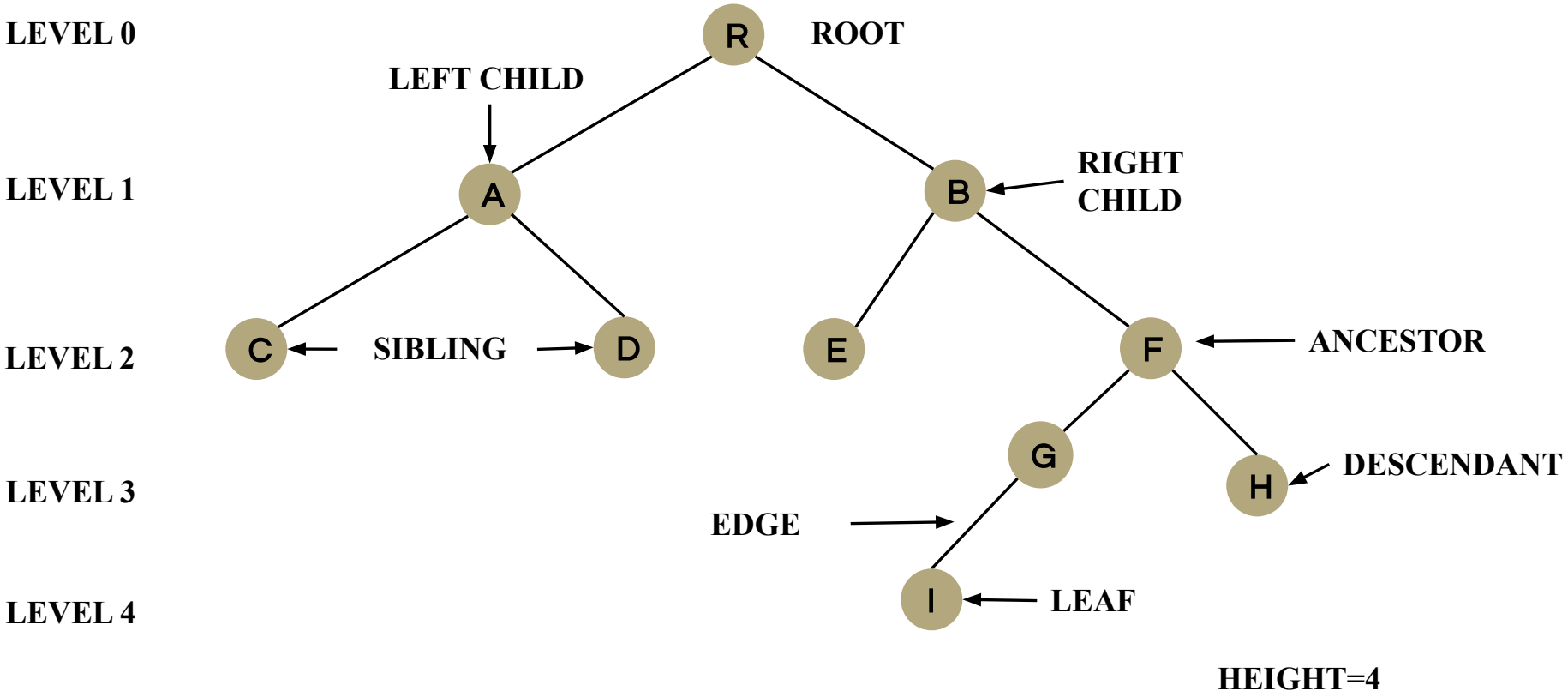
Binary tree



Properties of Tree

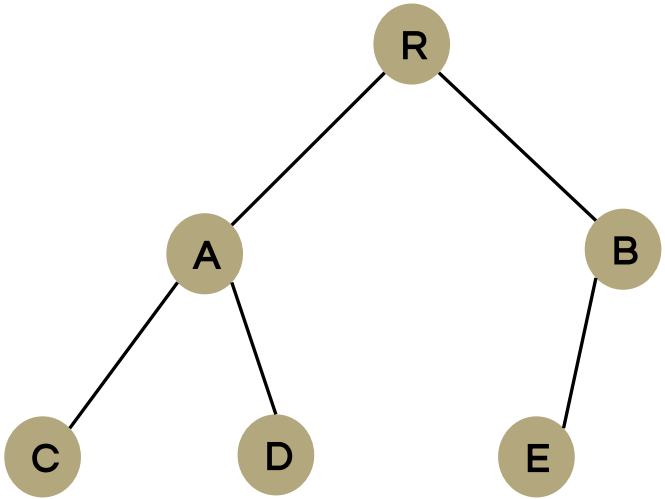
- A tree has exactly one path between every pair of vertices.
- A tree with n nodes always has n - 1 edges.
- A connected graph with n nodes and n - 1 edges is a tree.
- Trees are connected and acyclic.
- A tree is a minimally connected graph (removing any edge disconnects it).

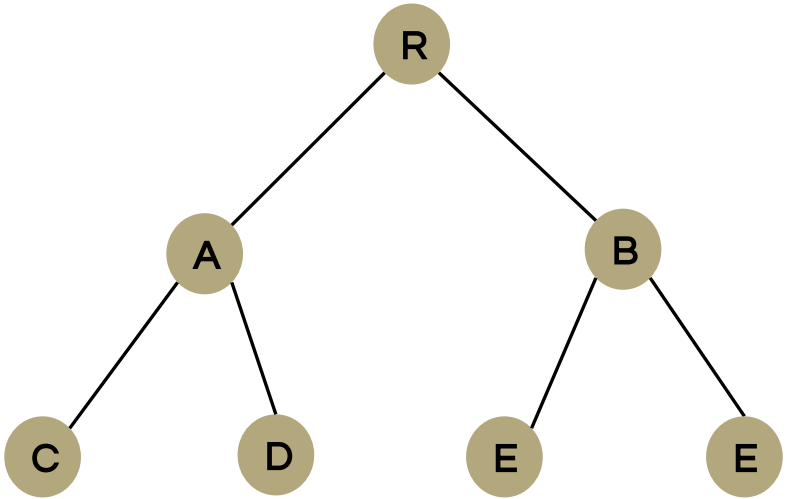
Trees Terminology



Binary Tree

- Each node has at most two children (left and right).
- A binary tree can be empty.
- Maximum degree of a node is two.
- Subtrees are ordered (left and right matter).

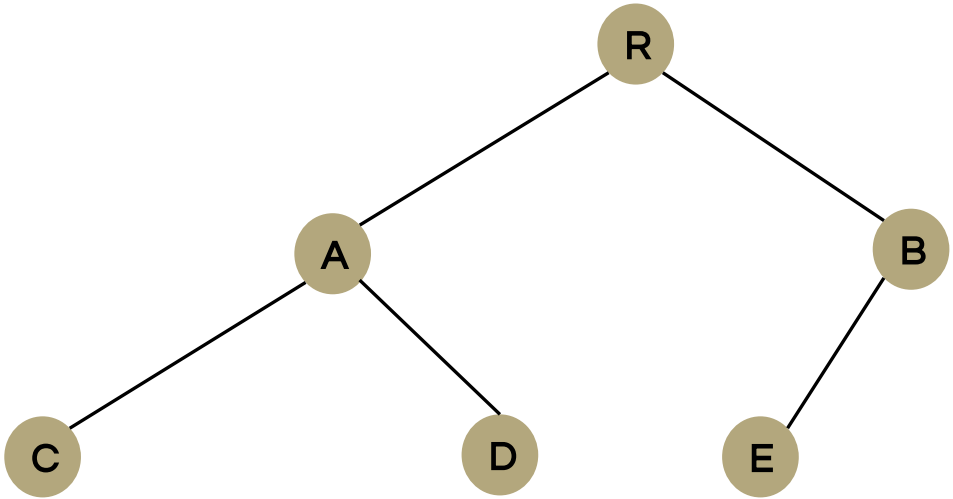




Complete Binary Tree

A Complete Binary Tree satisfies:

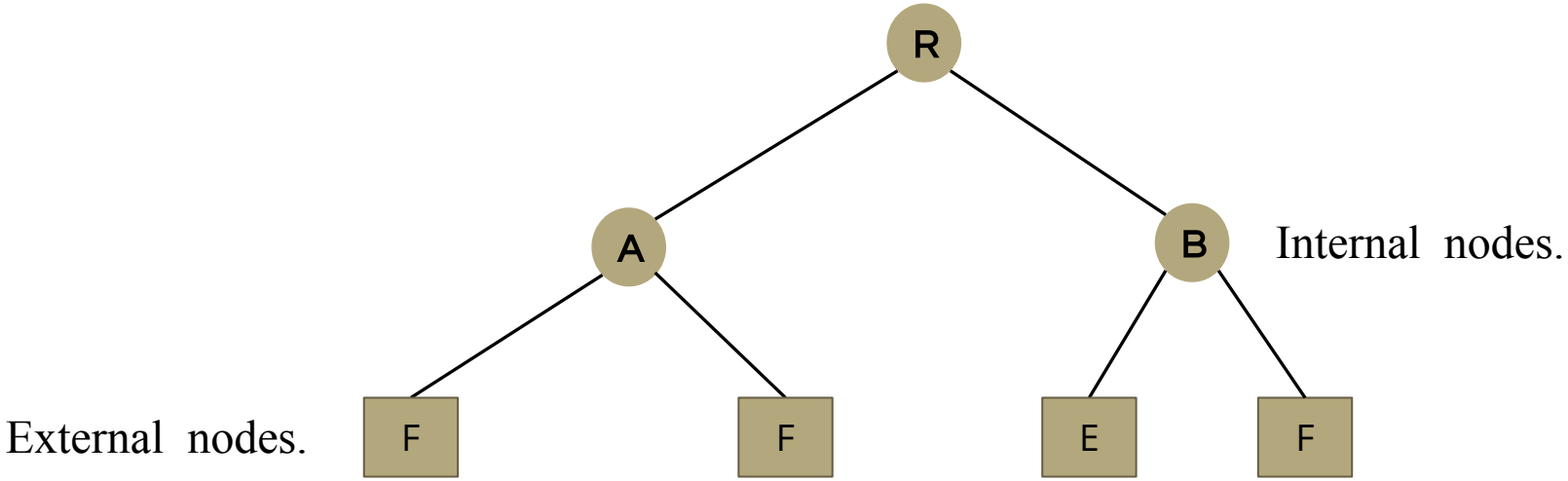
- All levels (except possibly the last) are completely filled.
- Nodes at the last level are filled from left to right without gaps.



Extended Binary Tree

An Extended Binary Tree (or 2-tree) is a binary tree where:

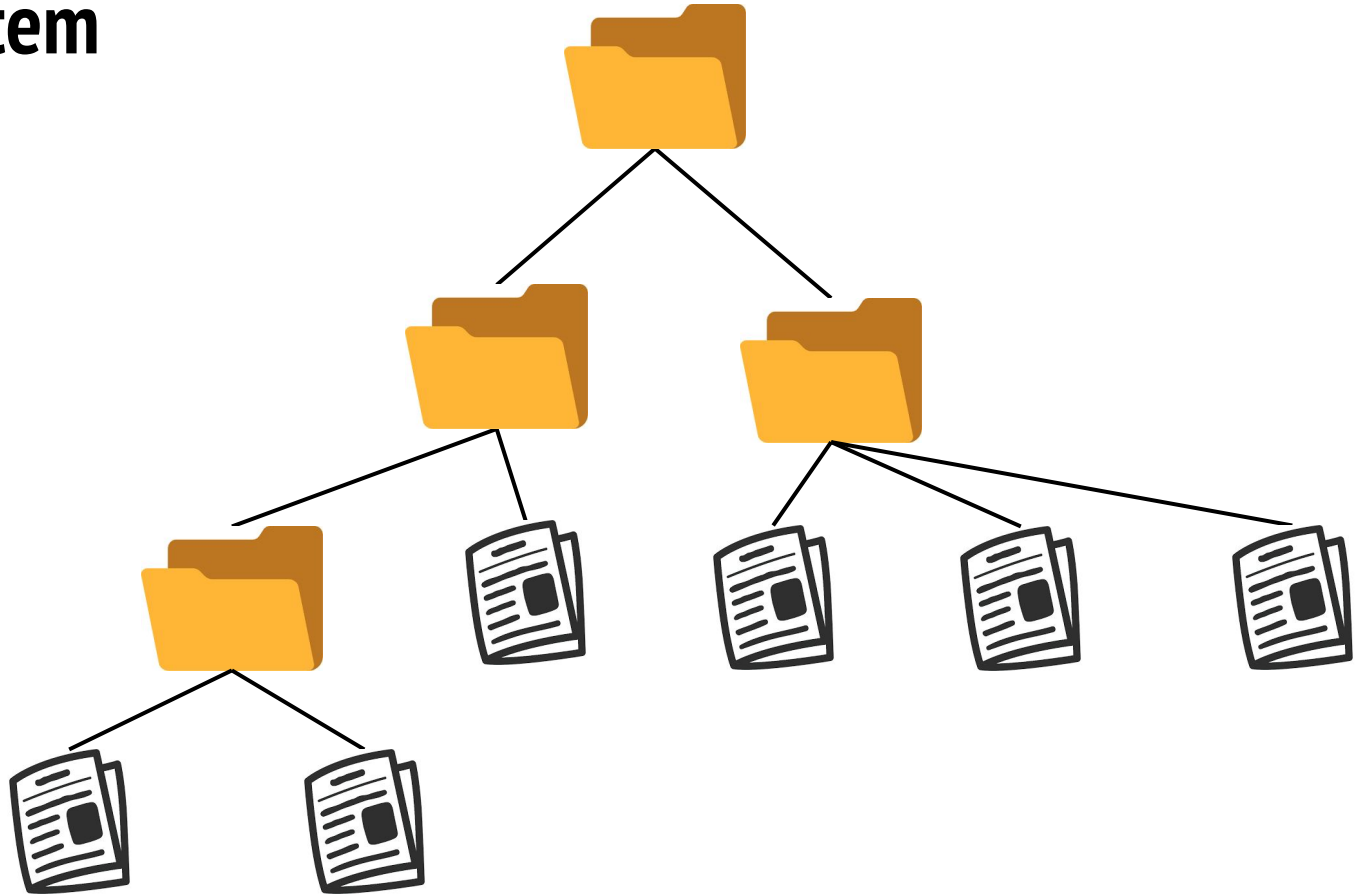
- Every node has either 0 or 2 children.
- Nodes with 2 children are called internal nodes.
- Nodes with 0 children are called external nodes (or leaves).



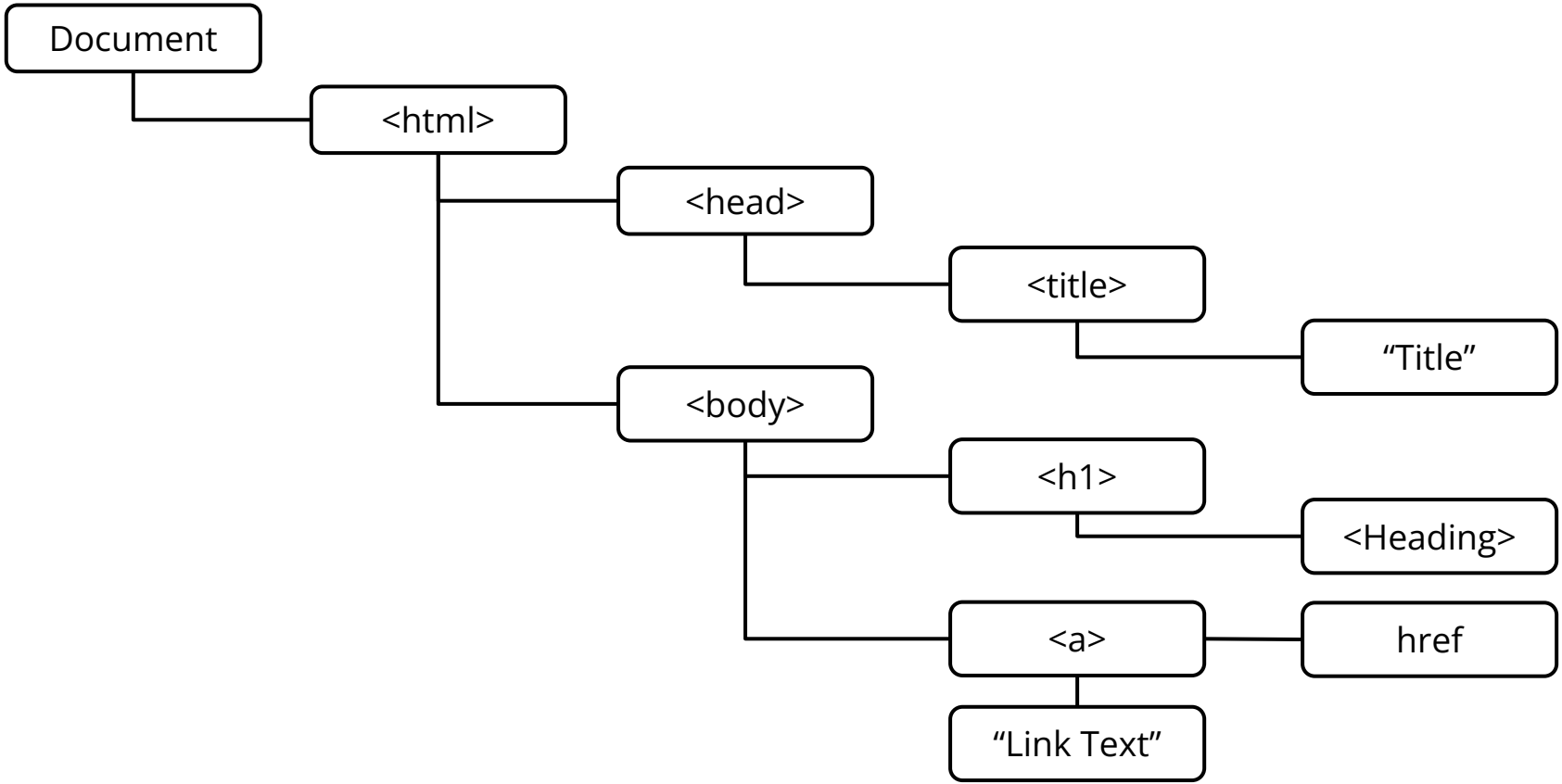


Applications of Trees

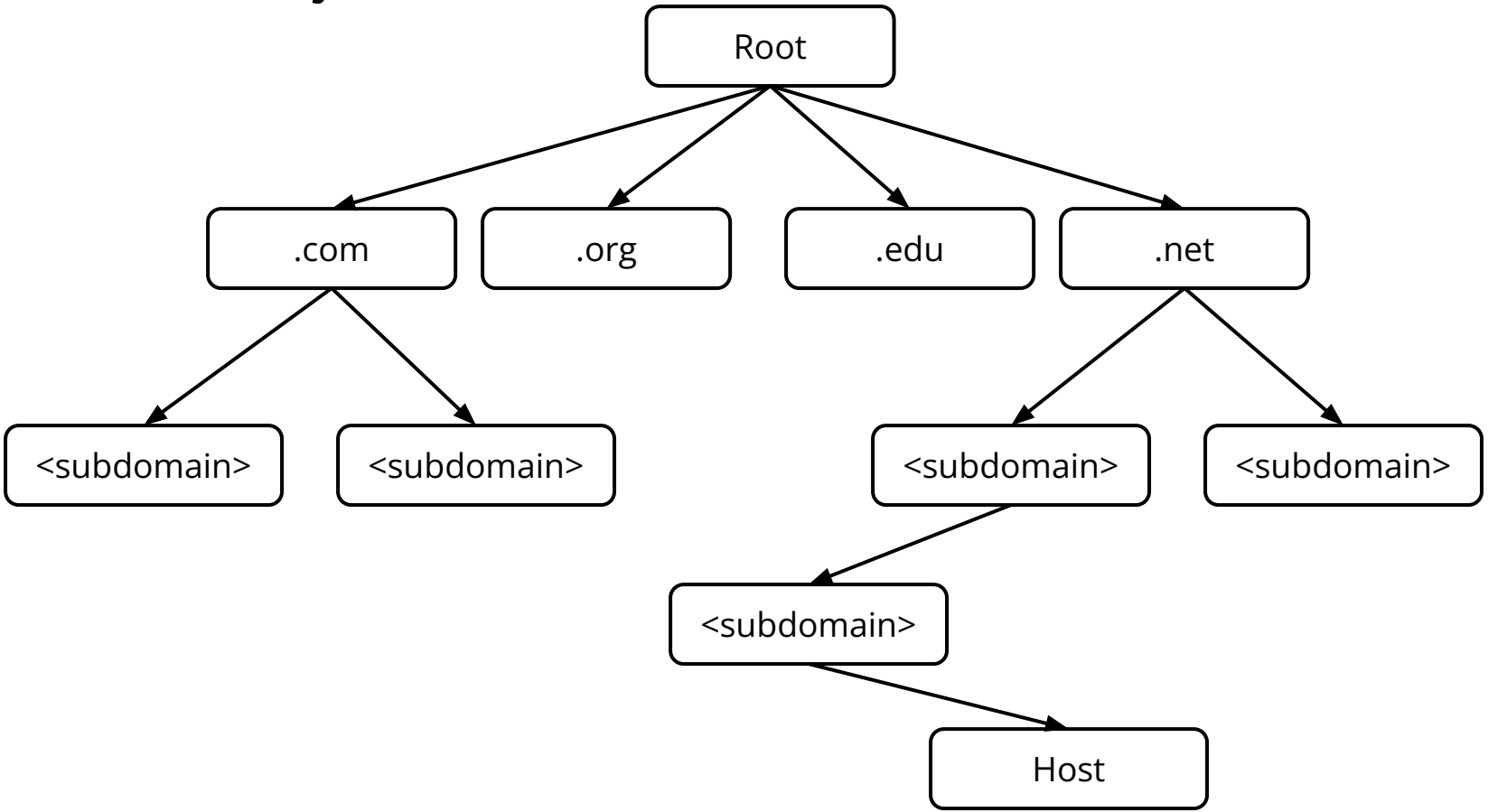
File System



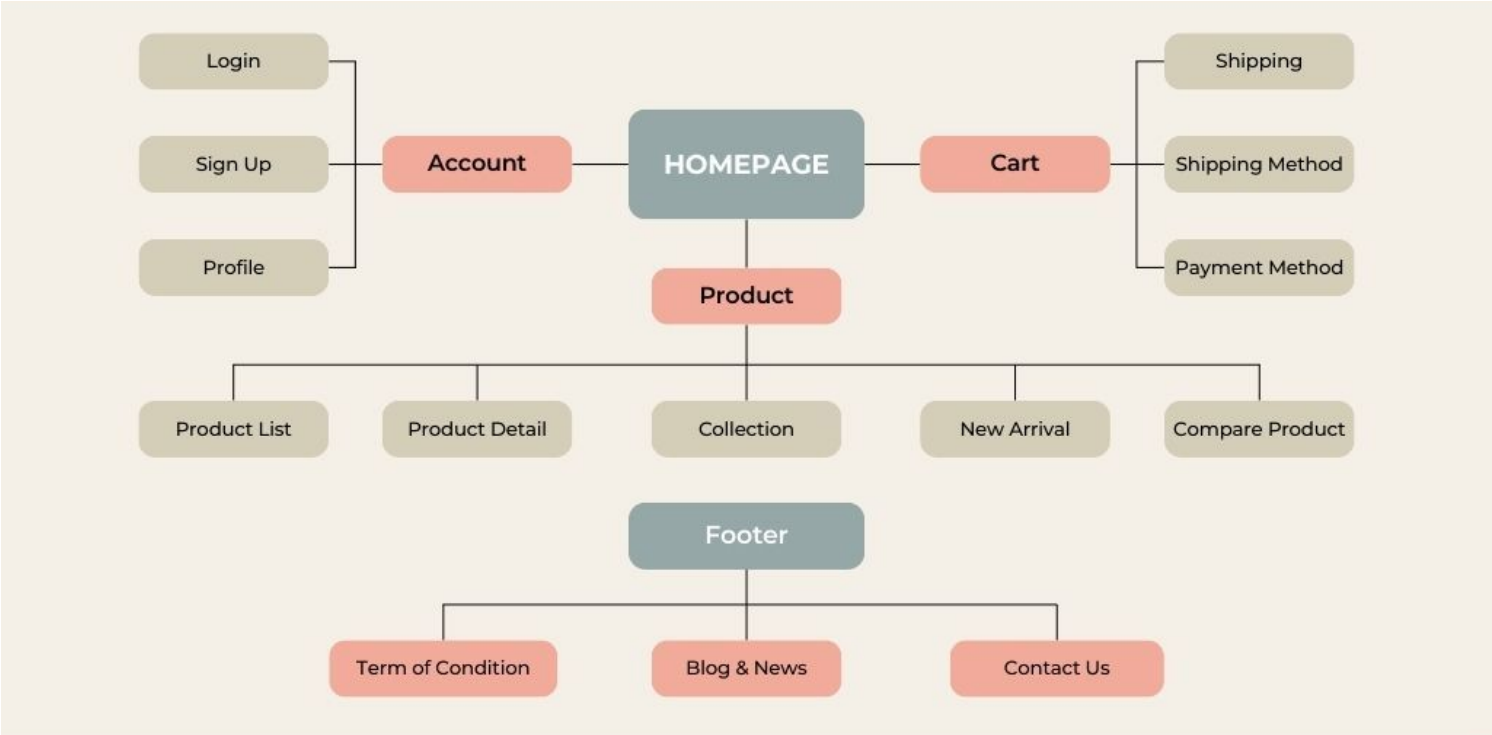
Document Object Modeling



DNS Hierarchy



Web Page Hierarchy

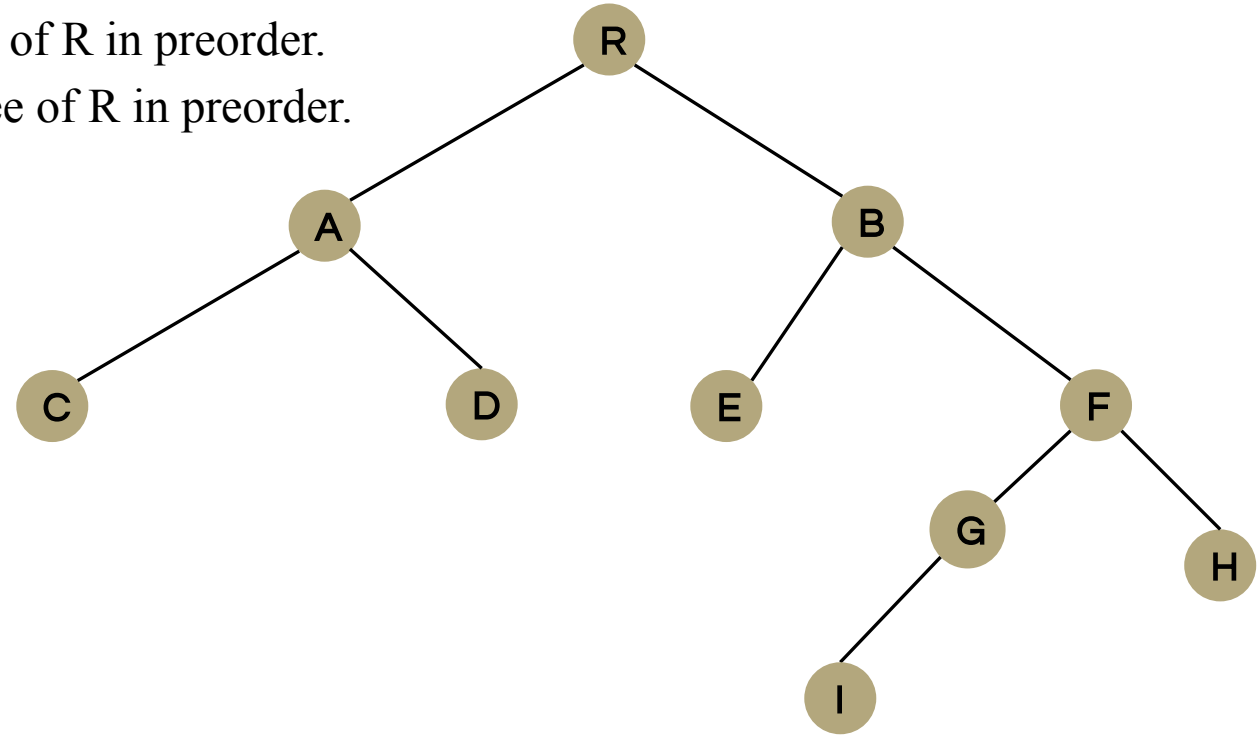




Traversal of Trees

Preorder Traversal of Binary Trees

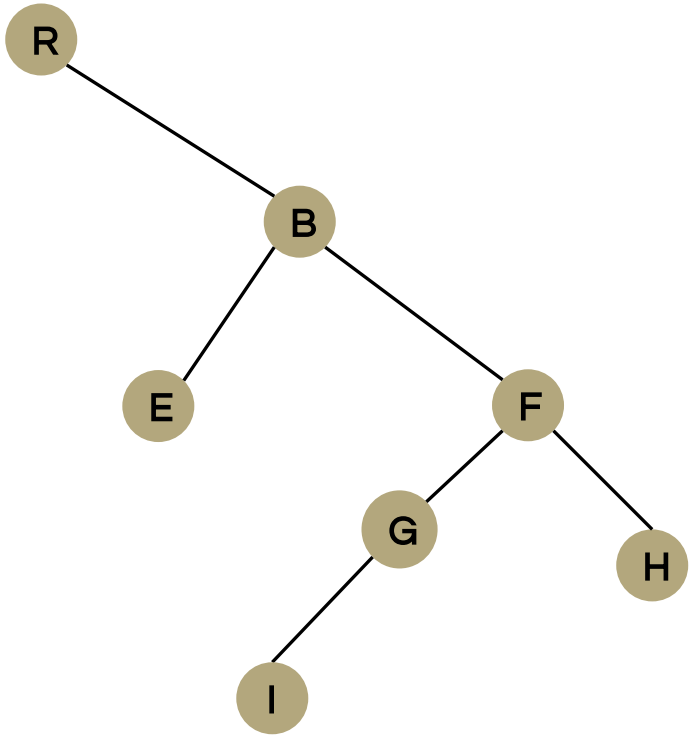
- Process the root R.
- Traverse thc left subtree of R in preorder.
- Traverse the right subtree of R in preorder.



R A C D B E F G I H

Preorder Traversal of Binary Trees

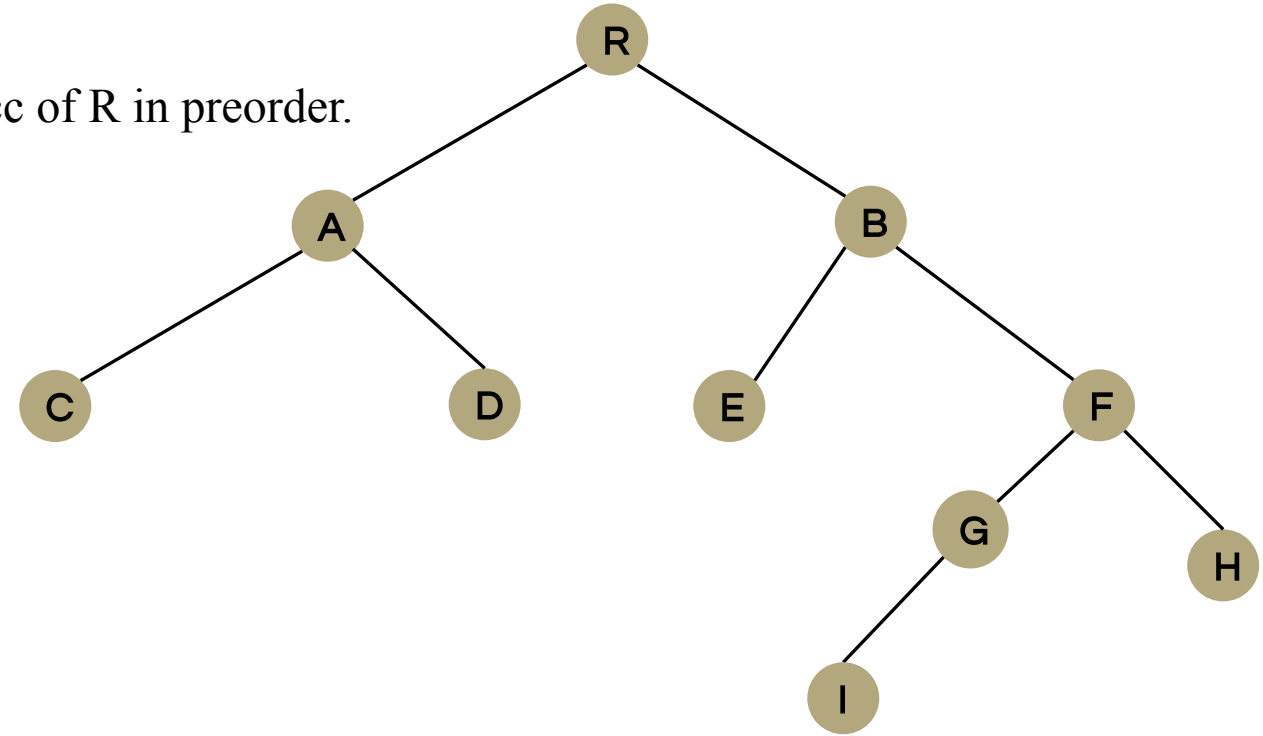
What is preorder traversal ?



R A C D B E F G I H

Inorder Traversal of Binary Trees

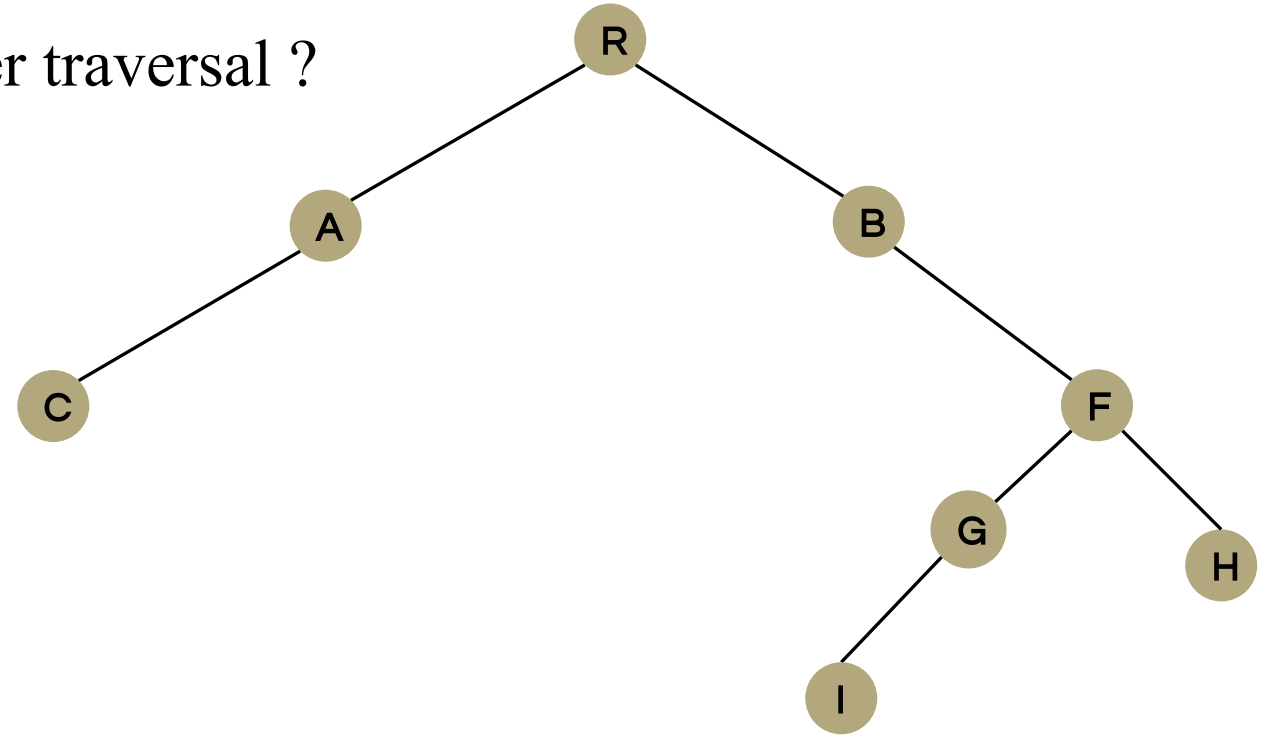
- Traverse thc left suhtrcc of R in preorder.
- Process the root R.
- Traverse the right subticc of R in preorder.



C A D R E B I G F H

Inorder Traversal of Binary Trees

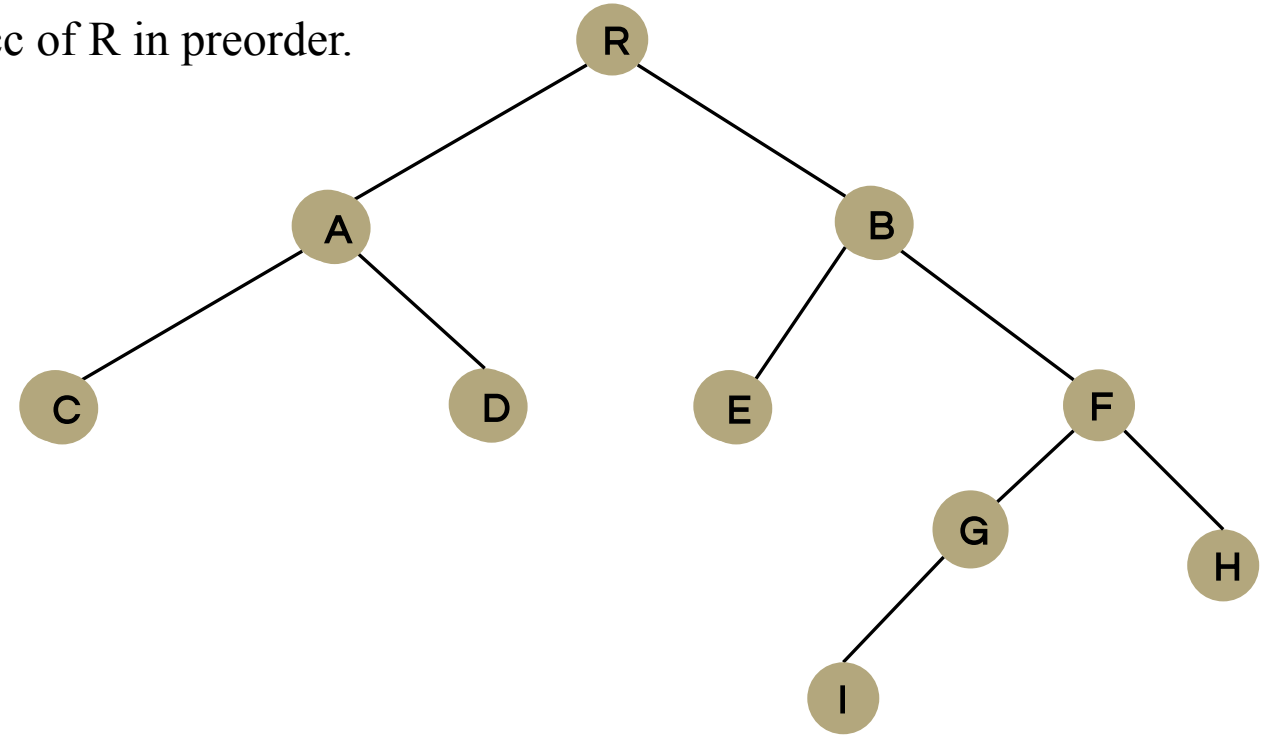
What is Inorder traversal ?



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Postorder Traversal of Binary Trees

- Traverse thc left suhtrcc of R in preorder.
- Traverse the right subticc of R in preorder.
- Process the root R.



C D A E I G H F B R

How to Implement Tree (Say Binary Tree)

Root

	Node	Left Child	Right Child
0	4	1	2
1	9	7	6
2	21	3	4
3	6	5	NULL
4	17	NULL	NULL
5	11	NULL	NULL
6	55	NULL	NULL
7	42	NULL	NULL

How to Implement Tree (Say Binary Tree)

